Employment & Unemployment Rate Analysis

A Data-Driven Analysis of Employment Rates by Country, Gender & Year

Presented by:

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Tools

SQL | Excel | PowerPoint

Introduction

Employment and unemployment rates are important indicators of a country's economic and social condition.
This project analyzes employment-related rate data across different countries, years and genders to identify patterns, differences and major changes over time.

Analysis Dimensions

- Country
- Male vs Female
- Year
- Rate
- Rate change

Problem Statement

The dataset contains employment rate information across countries, genders and years. However, raw data alone does not clearly show which countries perform better, how rates change over time, or whether there are significant differences between males and females.

- Which countries have the highest/lowest rates?
- How does the rate change over time?
- Is there a difference between Male and Female rates?
- Which countries have rates above the overall average?
- Which countries experienced the biggest changes?

Research Objectives

To analyze employment and unemployment rates across countries and over time, with a focus on gender differences, rate variations, and significant changes between the earliest and latest years.

Dataset Overview

Area

Country/Region

Type

Employment/Unemployment type

Source

Data source

Sex

Male / Female

Year

Year of observation

Rate

Recorded rate

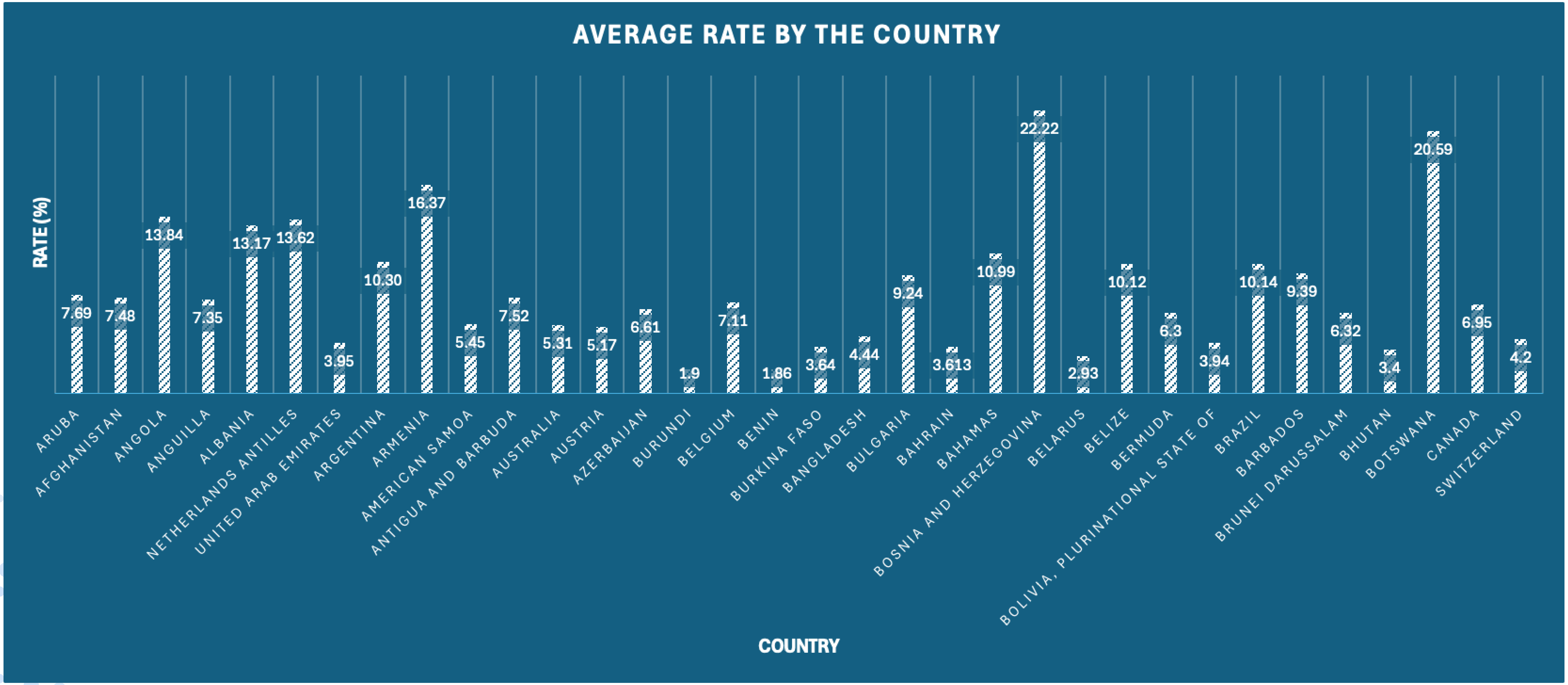
Data Analysis Methodology

- Raw Data
- Data Cleaning
- SQL Analysis
- Excel Visualization
- Findings & Insights

Tools Used

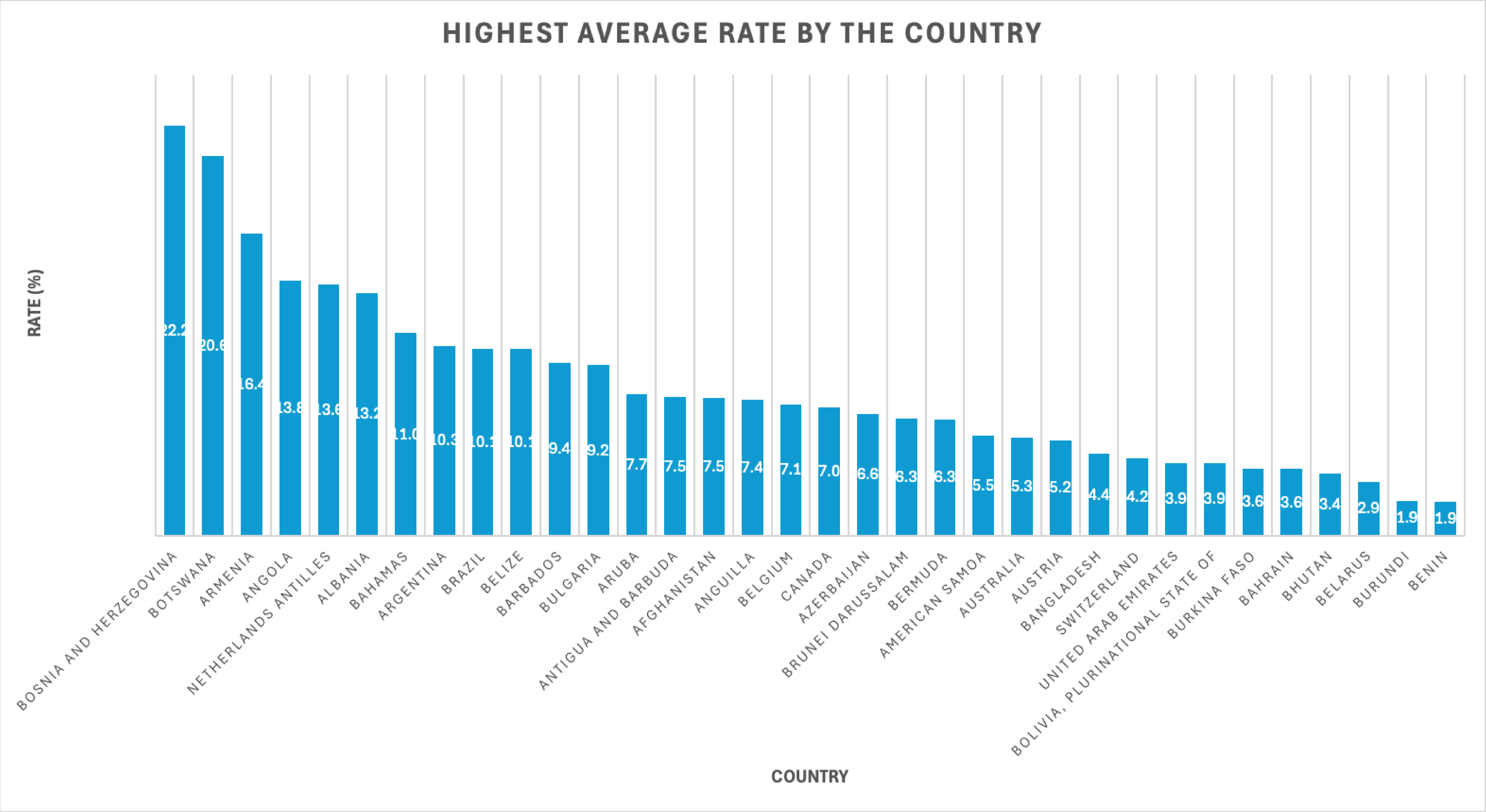
- SQL — Data analysis
- Excel — Charts & visualization
- PowerPoint — Presentation

Average Rate by Country



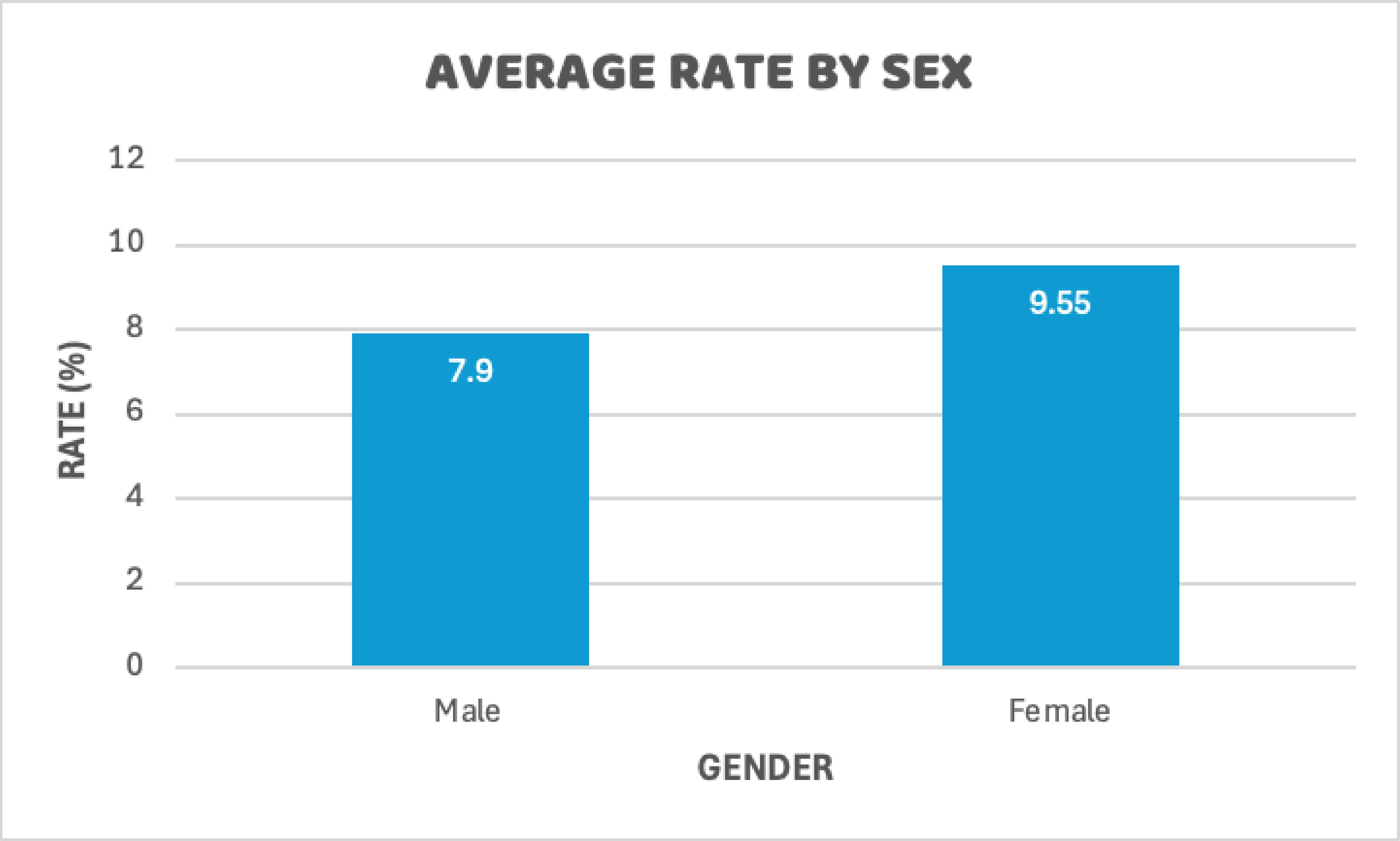
The analysis shows significant variation in average rates across countries.

Countries with Highest Average Rate



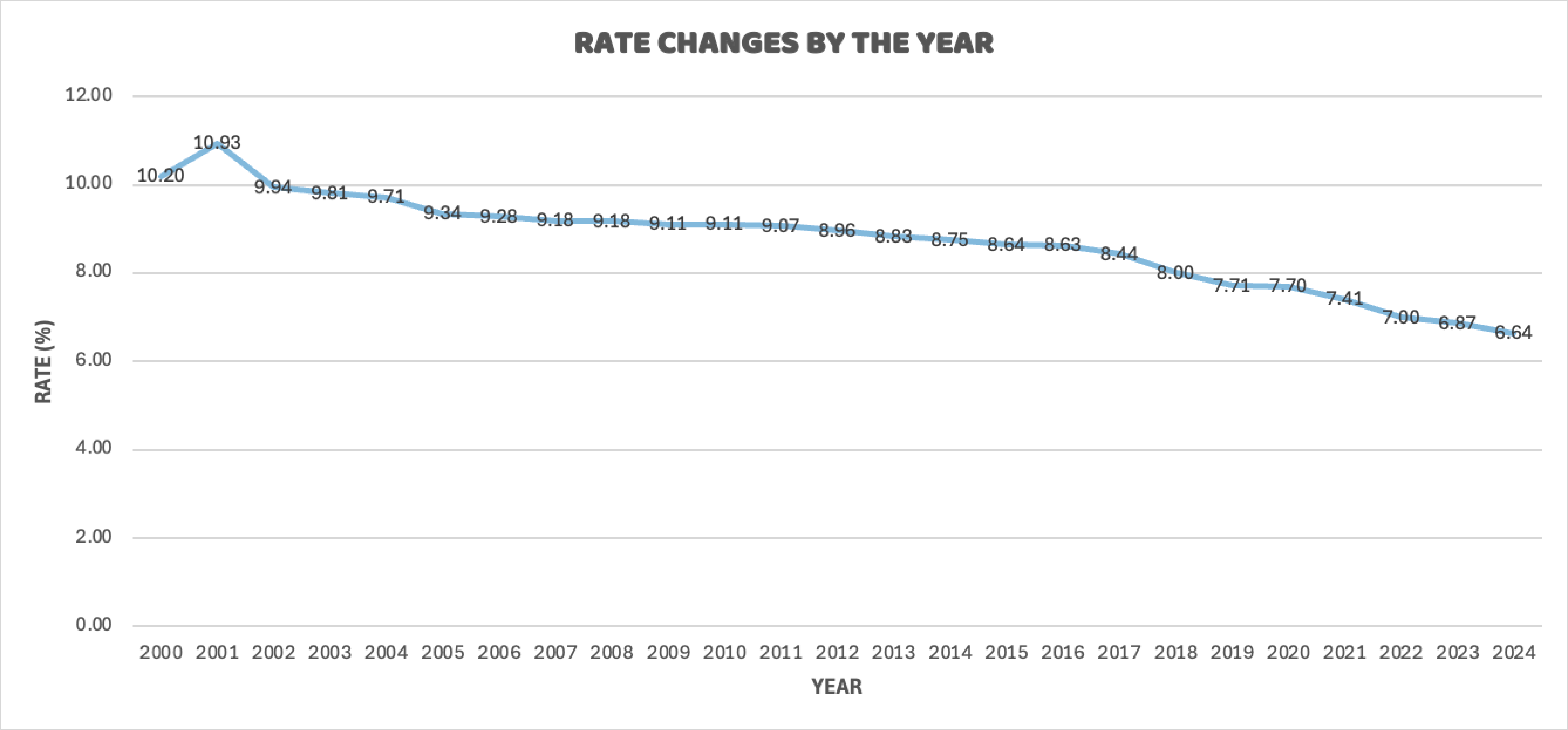
The top-performing countries show substantially higher average rates compared with other countries.

Male vs Female Analysis



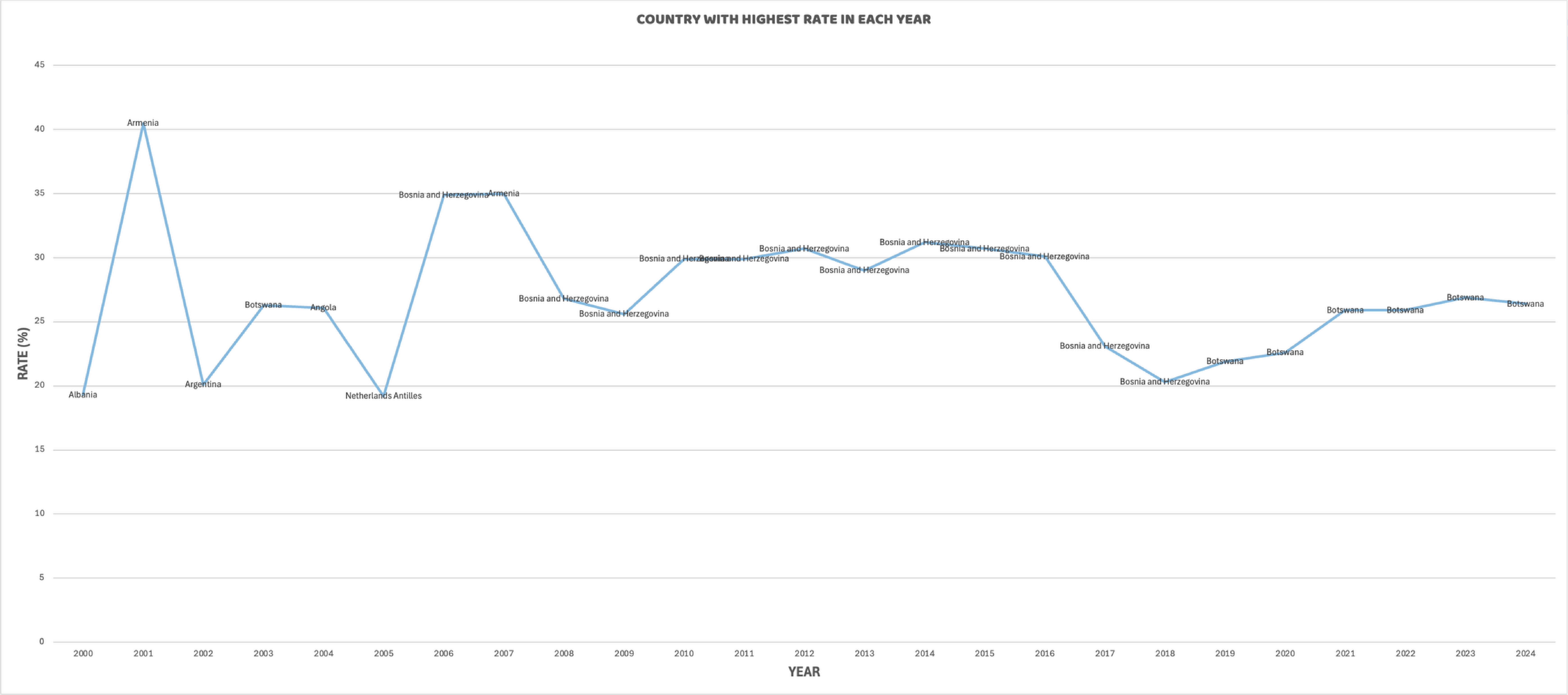
The comparison highlights the difference in average rates between males and females.

Rate Trend Over the Years



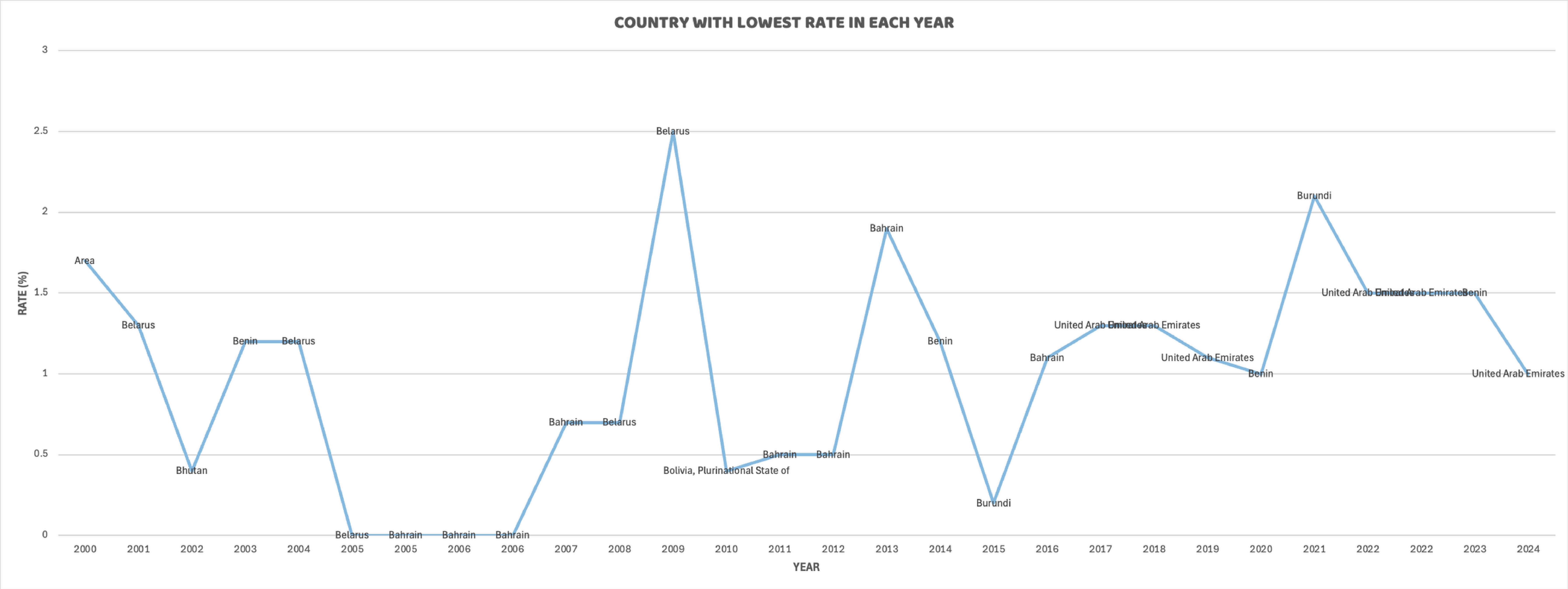
The yearly trend shows how the average rate has changed from 2000 to 2024.

Highest Rate in Each Year



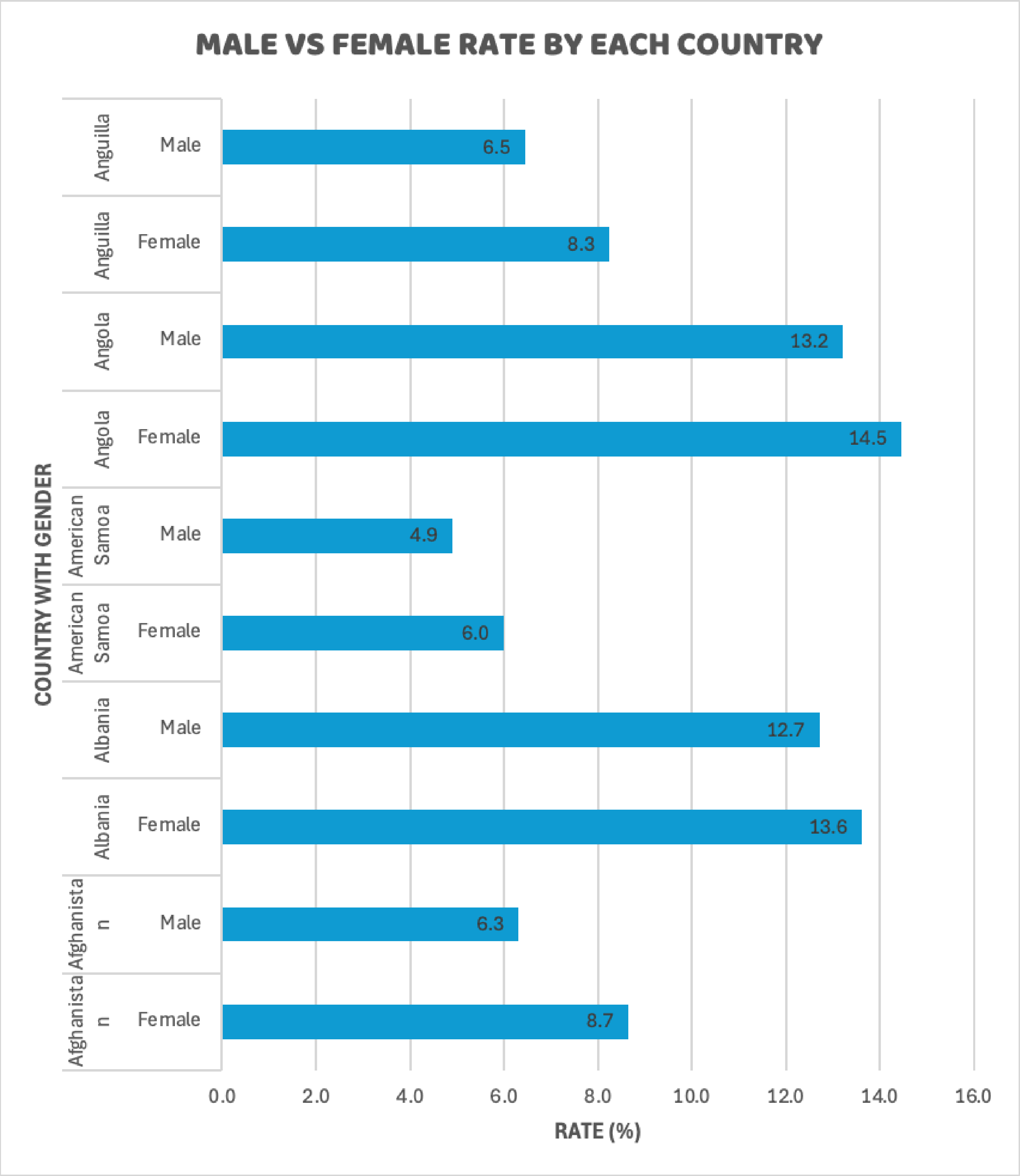
The highest rate declined over time and became relatively stable in recent years.

Lowest Rate in Each Year

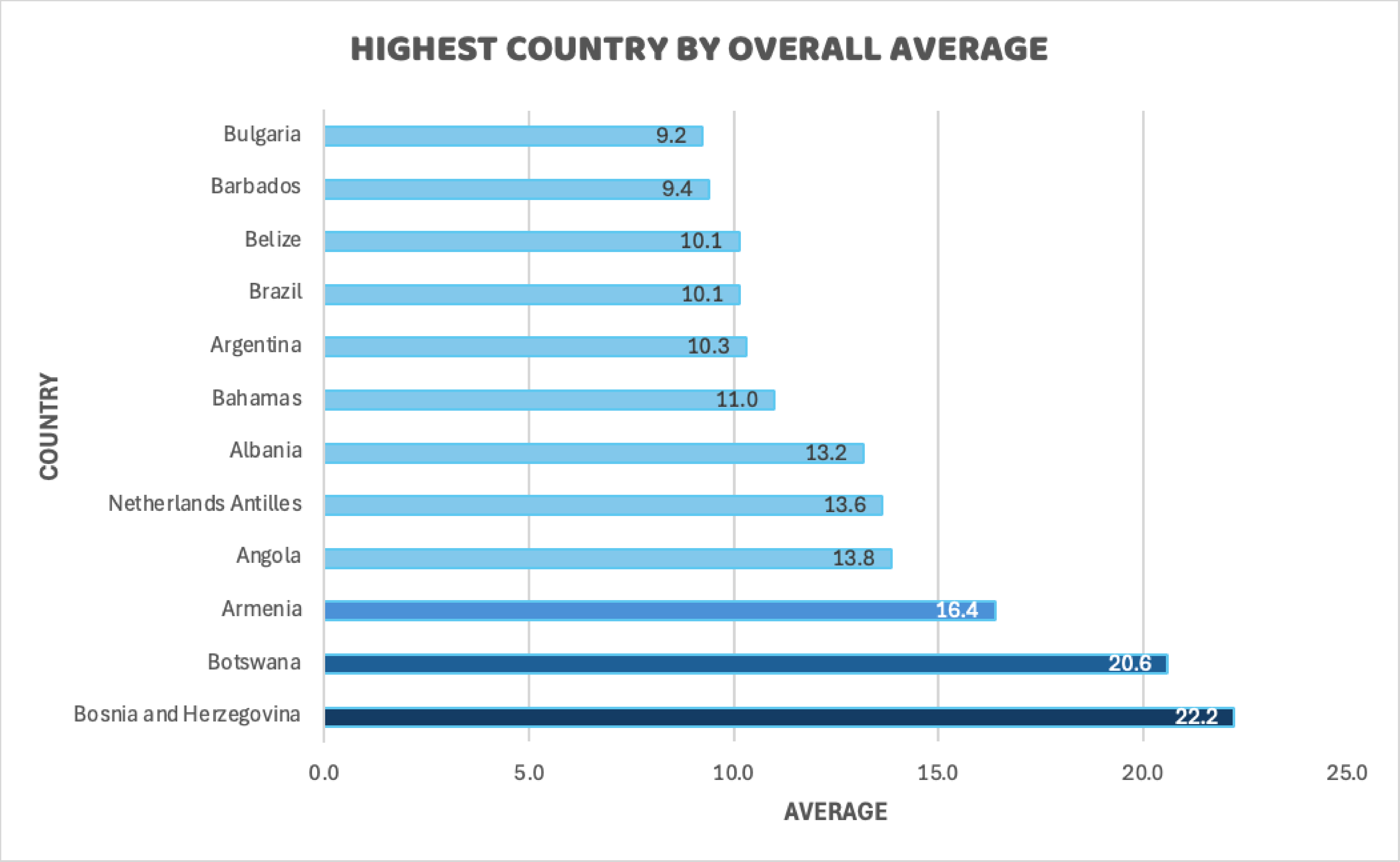


The lowest rate remained very low, with a slight increase in recent years.

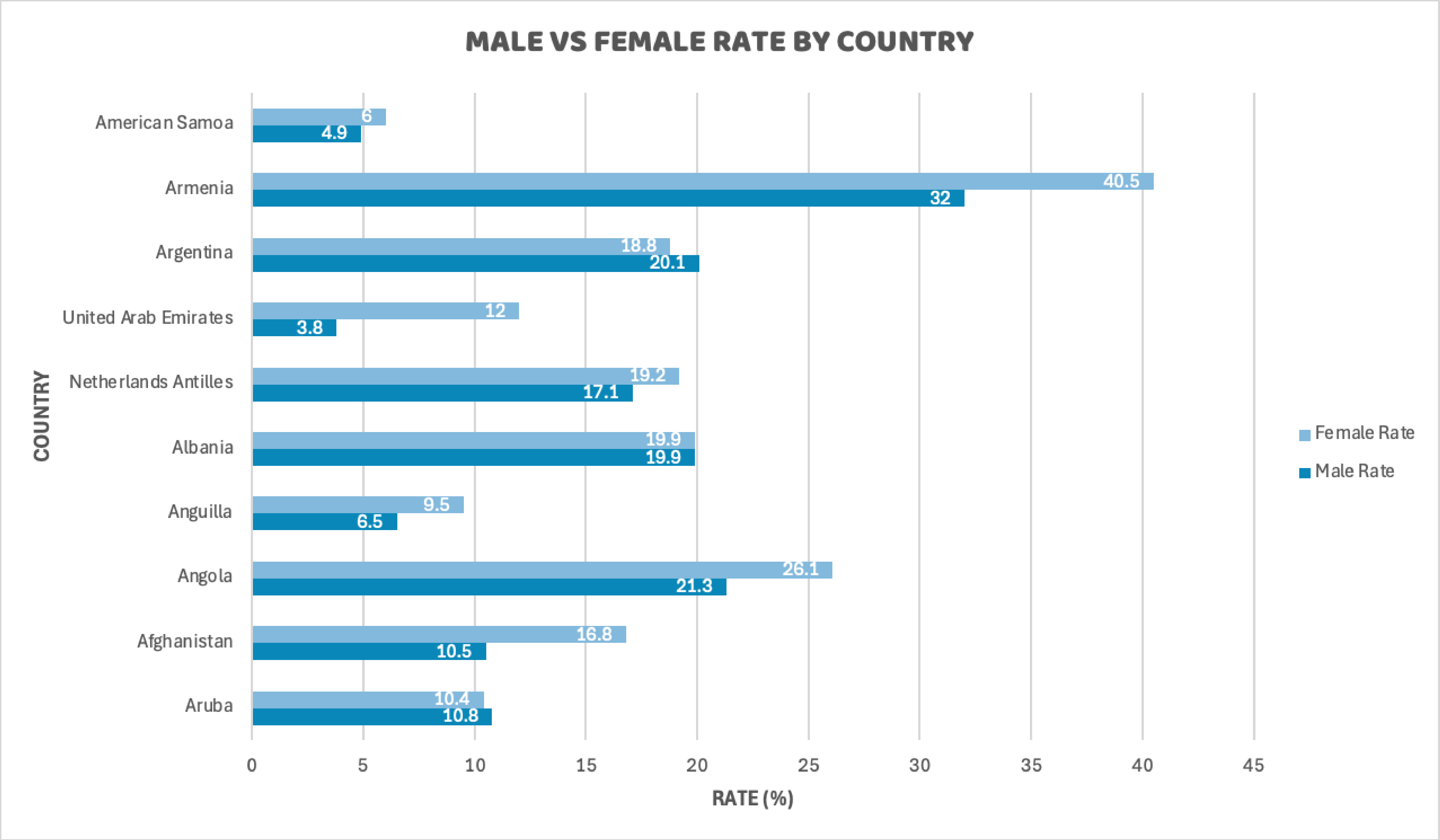
Top 10 Countries: Male vs Female



Countries Above Overall Average

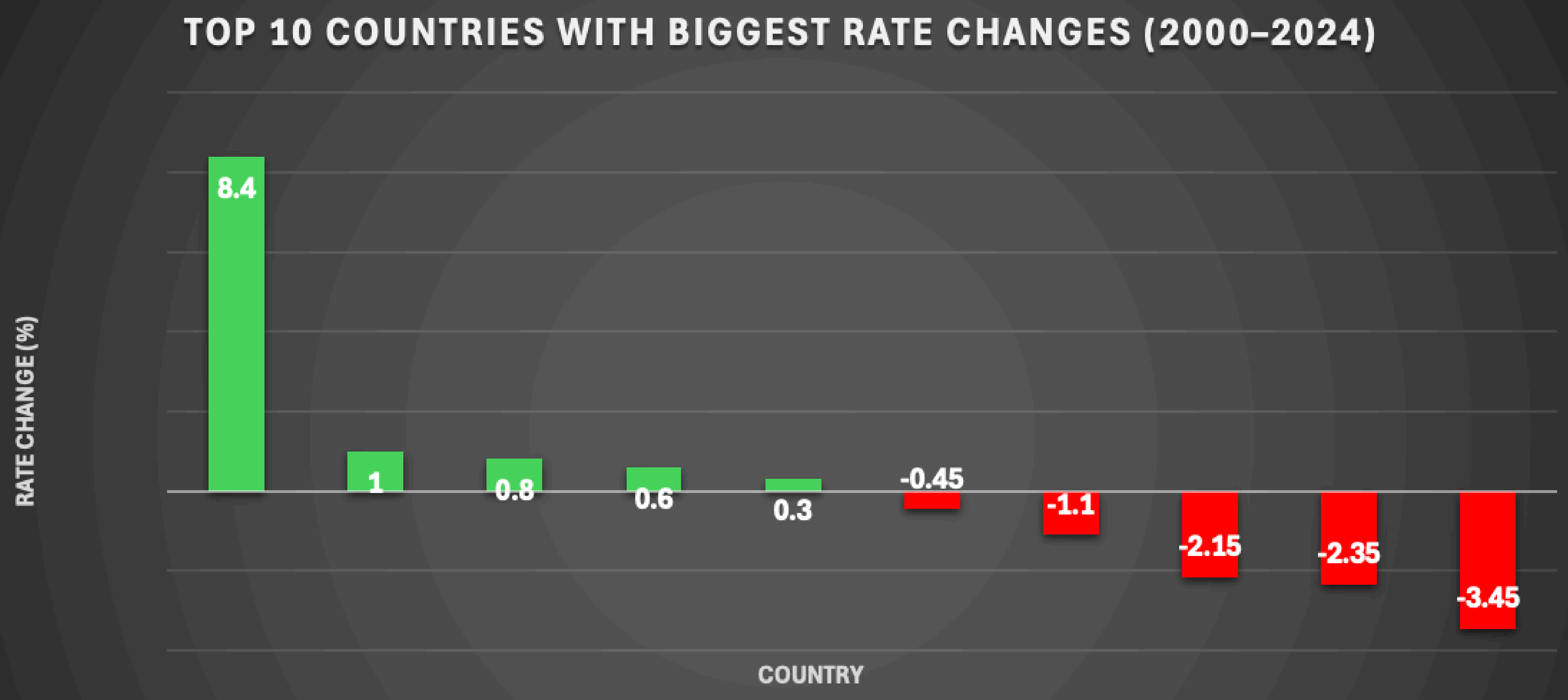


Gender Gap Analysis



The gender-gap analysis highlights countries where the difference between male and female rates is relatively large.

Biggest Rate Changes



	Botswana	Belarus	UAE	Austria	Bangladesh	Canada	Belgium	Bolivia	Australia	Barbados
Rate Change	8.4	1	0.8	0.6	0.3	-0.45	-1.1	-2.15	-2.35	-3.45

Key Findings

1	**Country Variation**	Average rates vary considerably across countries.
2	**Gender Difference**	Male and Female rates show noticeable differences in several countries.
3	**Time Trend**	The average rate changes over the 2000–2024 period.
4	**Highest & Lowest**	Different countries lead the highest and lowest rate rankings across years.
5	**Above Average Countries**	Several countries consistently perform above the overall average.
6	**Long-Term Change**	Some countries experienced substantial changes between 2000 and 2024.

Recommendations

- Countries with lower rates may require targeted employment policies.
- Gender differences should be monitored to identify employment inequalities.
- Long-term trends can help policymakers evaluate employment programs.
- Countries with significant rate changes should be studied further to understand the underlying causes.

Conclusion

This analysis provides a comprehensive view of employment-related rates across countries, genders and years. Using SQL and Excel, the project identifies country-level differences, gender gaps, yearly trends and significant changes over time. These insights demonstrate how data analysis can support better understanding of employment patterns.

Thank You

Questions & Discussion

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